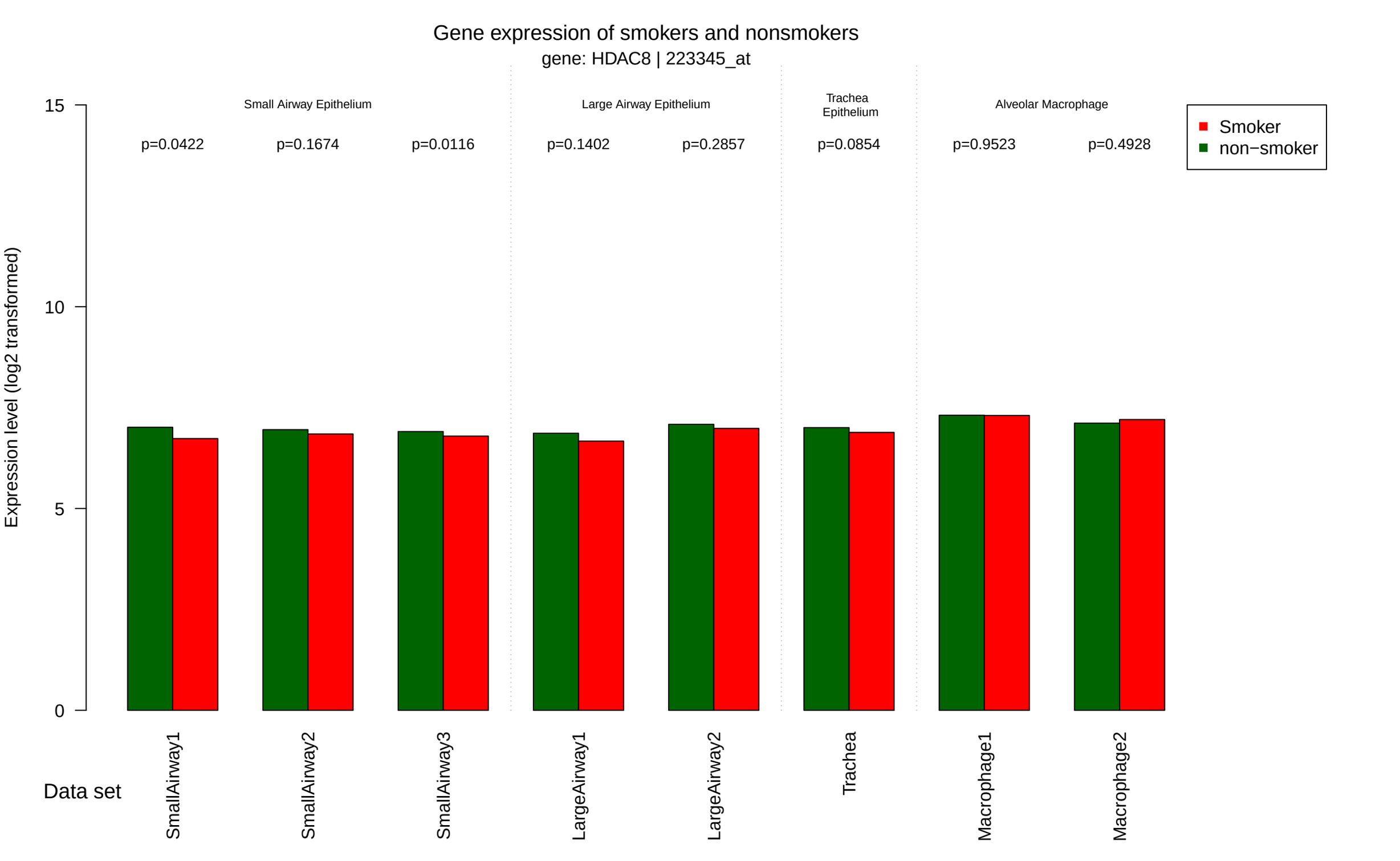
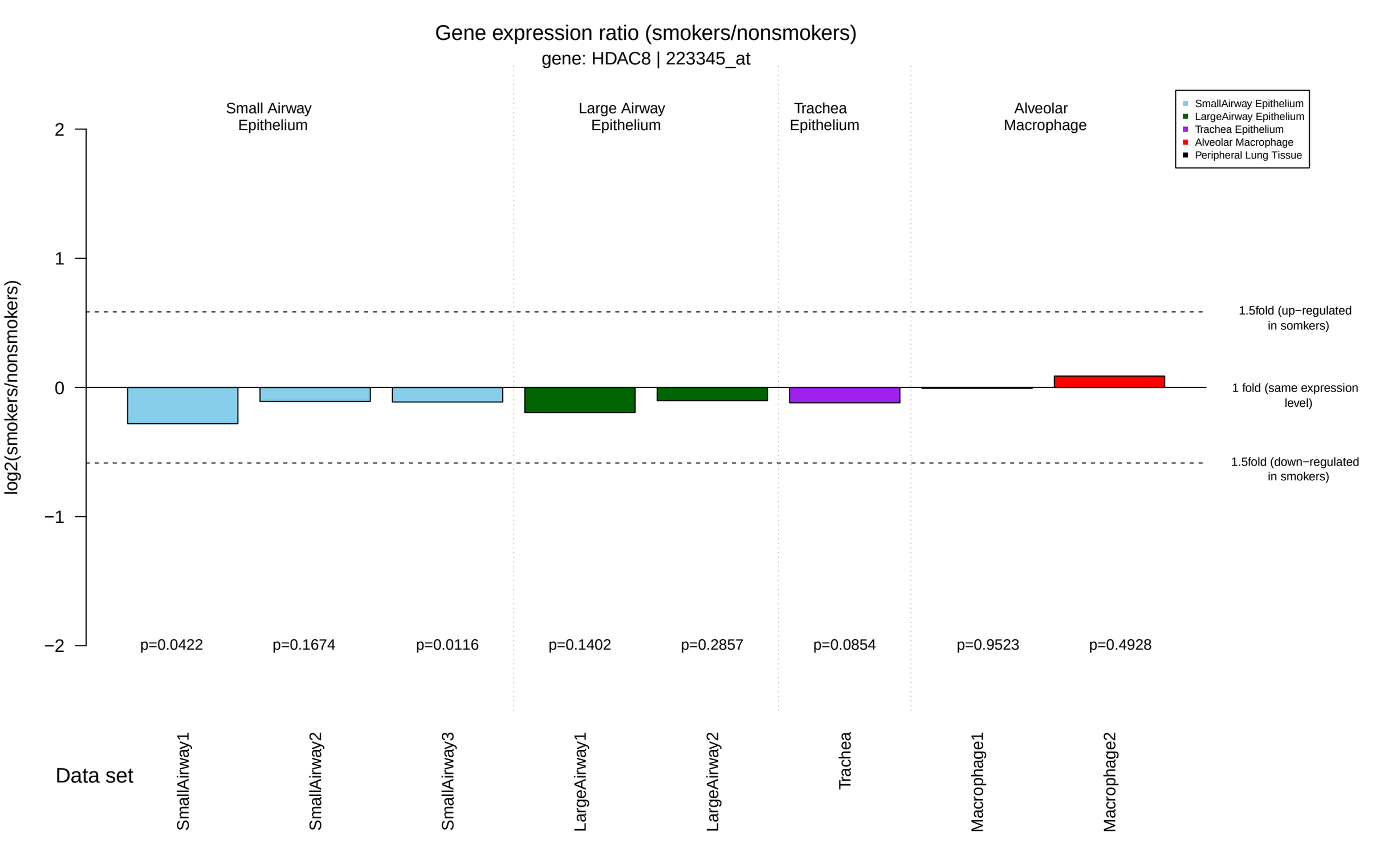
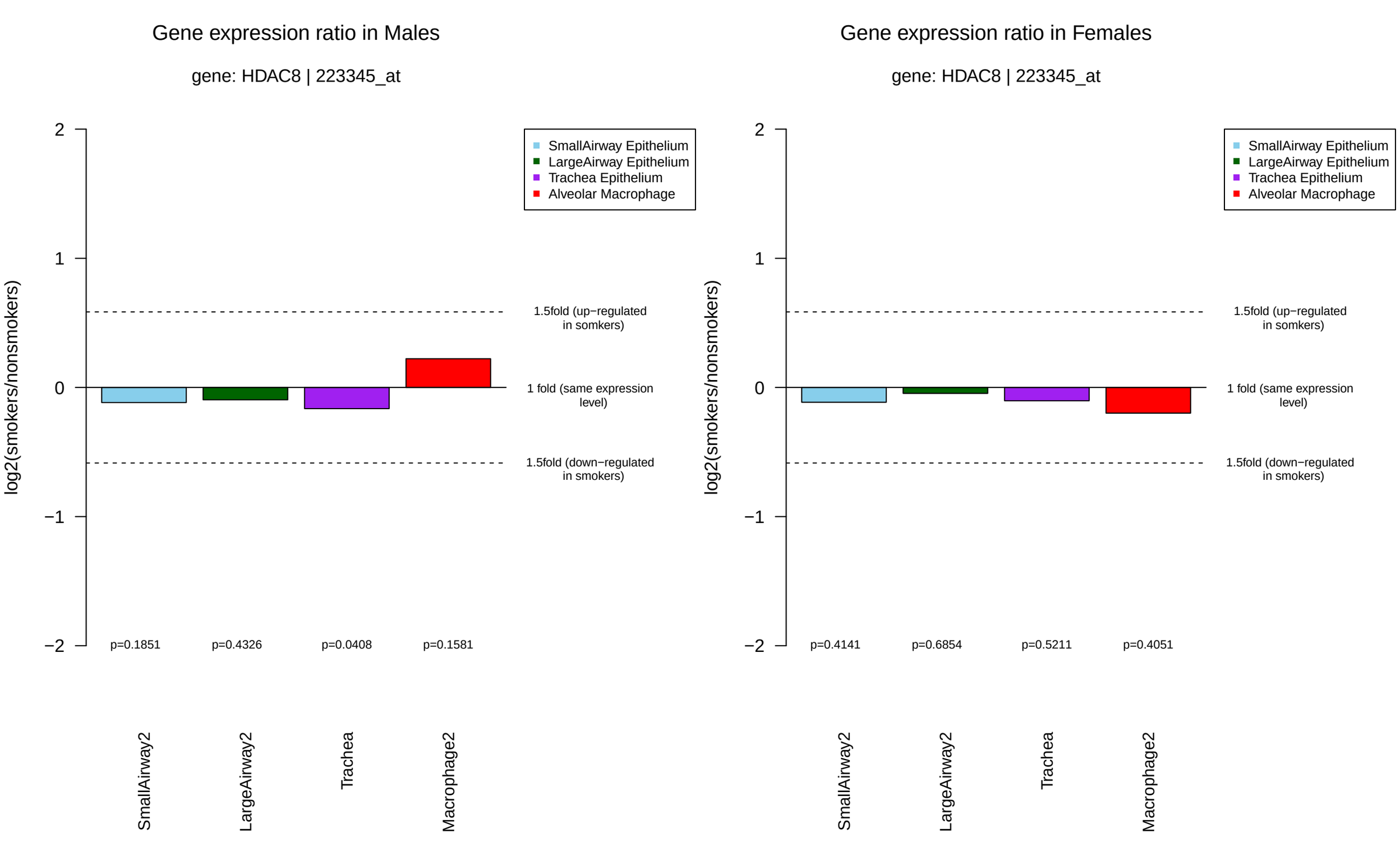
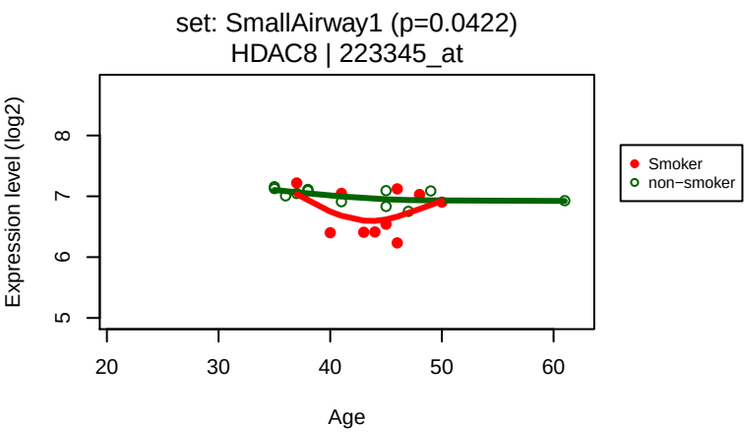
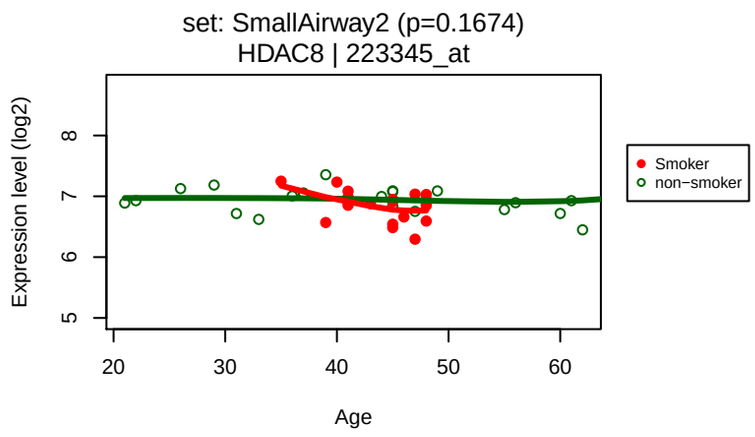
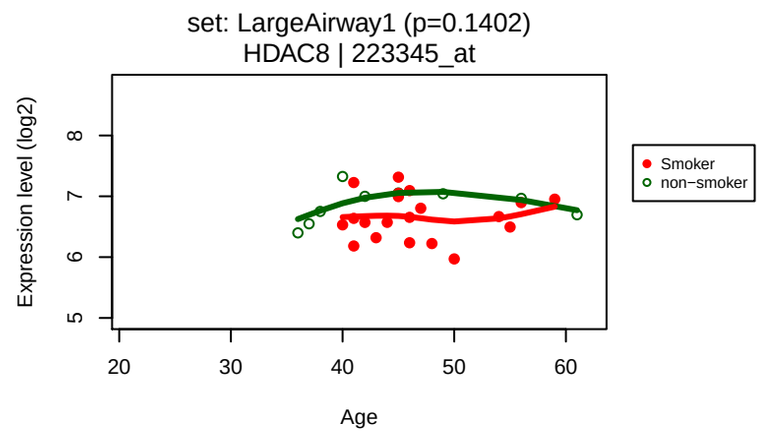

Gene expression of smokers and nonsmokers
gene: HDAC8 | 223345_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0422
p=0.1674
p=0.0116
p=0.1402
p=0.2857
p=0.0854
p=0.9523
p=0.4928
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

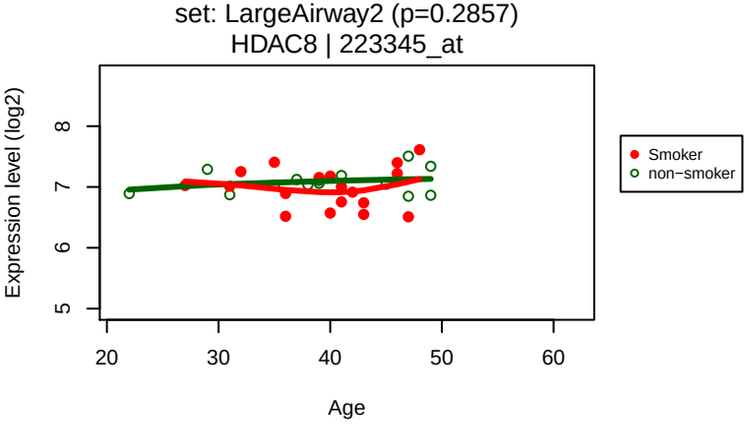

Gene expression ratio (smokers/nonsmokers)
gene: HDAC8 | 223345_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0422
p=0.1674
p=0.0116
p=0.1402
p=0.2857
p=0.0854
p=0.9523
p=0.4928
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

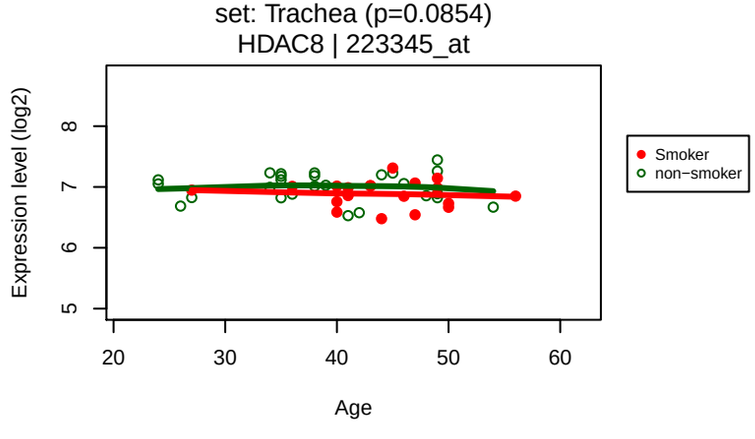

Gene expression ratio in Males
gene: HDAC8 | 223345_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1851
p=0.4326
p=0.0408
p=0.1581
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HDAC8 | 223345_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4141
p=0.6854
p=0.5211
p=0.4051
SmallAirway2
LargeAirway2
Trachea
Macrophage2

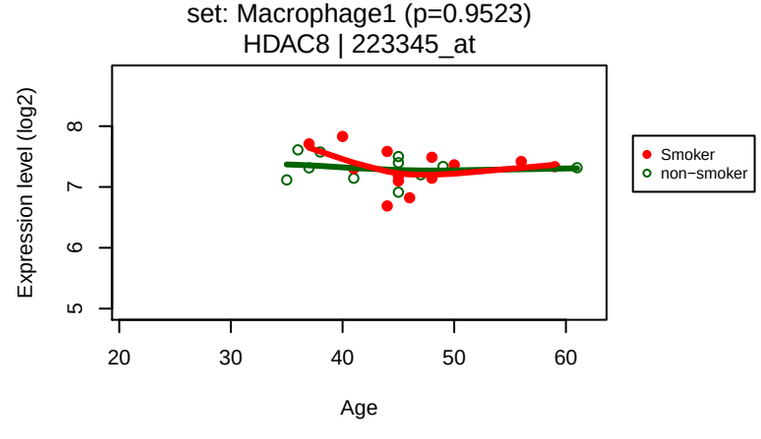

set: SmallAirway1 (p=0.0422)
HDAC8 | 223345_at
Expression level (log2)
Age
Smoker
non−smoker

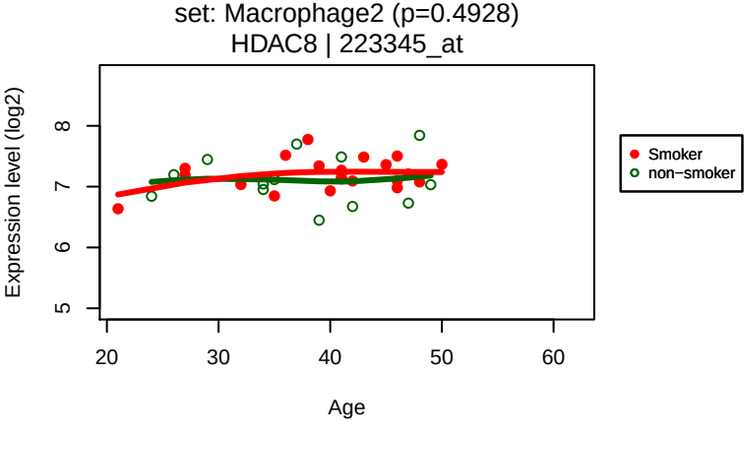

set: SmallAirway2 (p=0.1674)
HDAC8 | 223345_at
Expression level (log2)
Age
Smoker
non−smoker

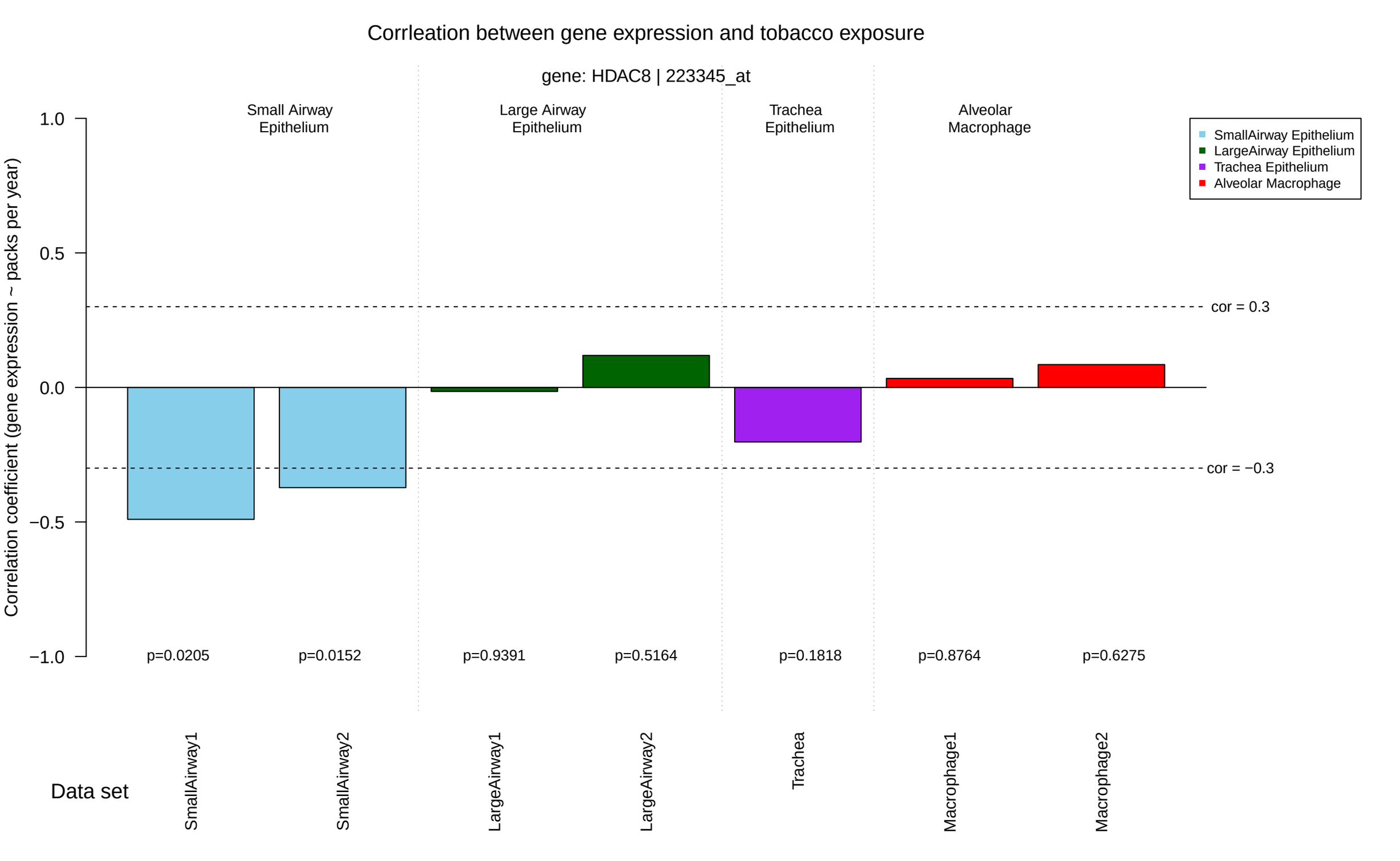

set: LargeAirway1 (p=0.1402)
HDAC8 | 223345_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2857)
HDAC8 | 223345_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0854)
HDAC8 | 223345_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9523)
HDAC8 | 223345_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4928)
HDAC8 | 223345_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HDAC8 | 223345_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0205
p=0.0152
p=0.9391
p=0.5164
p=0.1818
p=0.8764
p=0.6275
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2